

Software Requirements Specification

CocoConnect – A Digital Ecosystem for Coconut Estate Management

Version: 0.2

Status: Draft

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Client: Coconut Growers Association of Sri Lanka

1. Introduction

1.1 Purpose

This Software Requirements Specification describes the requirements identified for CocoConnect, a digital platform being developed to support coconut estate management and improve communication and business activities within the coconut industry.

The first version of the requirements focused mainly on identifying the problems faced by users and the main functions that the proposed system should provide. In this version, those requirements have been organized into separate modules to provide a clearer understanding of the system structure.

The requirements in this document are still subject to change based on further stakeholder discussions and analysis.

1.2 Scope

CocoConnect will provide a platform where different stakeholders involved in the coconut industry can manage their activities through a single system.

The system will support user management, estate management, product buying and selling, workforce and service provider management, harvest recording, ratings and reviews, notifications and administration.

The initial focus is on the coconut industry. However, the system should be flexible enough to support additional agricultural products and services in future versions.

2. Overall Description

2.1 Product Perspective

CocoConnect is planned as a mobile and web based application. The system will connect coconut estate owners, estate managers, buyers, sellers, workers and service providers through a common platform.

The system will contain several modules, with each module handling a specific area of functionality. These modules will work together to provide the overall CocoConnect experience.

The proposed technology stack consists of Flutter for the frontend, Node.js and Express.js for backend development and PostgreSQL for database management.

2.2 User Roles

The main user roles identified for the system are Estate Owner, Estate Manager, Buyer, Seller, Worker or Service Provider and Administrator.

Estate Owner

An Estate Owner is responsible for managing one or more coconut estates. The owner can maintain estate information, monitor harvest activities, find workers and service providers and participate in product selling activities.

Estate Manager

An Estate Manager is responsible for the daily activities of an assigned estate. The manager can coordinate workers, manage estate activities and record harvest information.

Buyer

A Buyer can browse available products, search for products, view seller information and communicate with sellers.

Seller

A Seller can create and manage product listings and communicate with buyers.

Worker or Service Provider

A Worker or Service Provider provides labour, machinery or other services required by estate owners.

Administrator

An Administrator is responsible for managing users, monitoring platform activities and maintaining the overall integrity of the system.

3. System Modules

3.1 User Management Module

Module Definition

The User Management Module is responsible for managing user accounts and controlling access to the system.

It handles activities such as registration, login, profile management and role based access. This module ensures that users can access the functions relevant to their assigned role.

Main Functions

User registration

User login

Password recovery

Profile management

Profile picture management

Role assignment

Role based access

Logout

Main Users

Estate Owner

Estate Manager

Buyer

Seller

Worker or Service Provider

Administrator

3.2 Product Marketplace Module

Module Definition

The Product Marketplace Module allows users to advertise and search for agricultural products. Sellers can create product listings while buyers can browse available products and contact sellers.

The initial focus of this module is on coconut related products.

Main Functions

Create product listings

Edit product listings

Remove product listings

Browse products

Search products

Filter products

View product details

Manage product availability

Contact sellers

Initial Product Categories

DC Coconut

Curry Coconut

Coconut Husk

Other agricultural products that may be introduced later

Main Users

Buyer

Seller

Estate Owner when acting as a seller

3.3 Workforce Marketplace Module

Module Definition

The Workforce Marketplace Module connects estate owners with workers and service providers who can provide services required for estate operations.

Workers and service providers can create profiles describing the services they provide, while estate owners can search for suitable providers and send job requests.

Main Functions

Create service profiles

Search workers and service providers

View service information

Send job requests

Accept job requests

Reject job requests

Track job request status

Initial Service Categories

Coconut Plucking

Husking

Fertilizing

Coconut Gathering

Coconut Loading

Tractor Services

Equipment Supply

Charging Methods

Per Tree

Per Day

Main Users

Estate Owner

Worker

Service Provider

3.4 Estate Management Module

Module Definition

The Estate Management Module allows estate owners to register and maintain information about their coconut estates.

The module provides a central place to store estate information and allows authorized users to view and manage estate related information.

Main Functions

Register an estate

Update estate information

View estate information

Manage assigned estate information

View estate related statistics

Main Users

Estate Owner

Estate Manager

3.5 Harvest Management Module

Module Definition

The Harvest Management Module is used to record and maintain information about coconut harvests.

Estate owners and authorized estate managers can record harvest information for the relevant estate. The stored information can later be used to review previous harvests and support reporting.

Main Functions

Record harvest

Update harvest information

View harvest records

View harvest history

Maintain historical harvest data

Generate basic harvest information for reporting

Main Users

Estate Owner

Estate Manager

3.6 Rating and Review Module

Module Definition

The Rating and Review Module is intended to help users build trust when interacting with each other.

Users can provide ratings and written reviews after completing a transaction or service. The ratings can help other users understand the reliability of sellers, buyers, workers and service providers.

Main Functions

Give ratings

Write reviews

View ratings

View reviews

Calculate average ratings

Main Users

Estate Owner

Estate Manager

Buyer

Seller

Worker

Service Provider

3.7 Notification Module

Module Definition

The Notification Module provides users with updates about important activities taking place within the platform.

Notifications can be related to job requests, user interactions, account activities and other events that require the user's attention.

Main Functions

Job request notifications

Request acceptance notifications

Request rejection notifications

Seller contact notifications

Account activity notifications

System notifications

Main Users

All registered users

3.8 Administration Module

Module Definition

The Administration Module allows administrators to monitor and manage the CocoConnect platform.

The module provides administrative functions for managing users, monitoring product listings and handling reported content.

Main Functions

View users

Manage user accounts

Verify users

Monitor product listings

Review reported content

Suspend users

Remove inappropriate listings

View platform reports

Main Users

Administrator

4. Functional Requirements

4.1 User Management Requirements

FR01: The system shall allow users to register an account.

FR02: The system shall allow users to log in using their registered credentials.

FR03: The system shall allow users to reset forgotten passwords.

FR04: The system shall allow users to update their profile information.

FR05: The system shall allow users to upload a profile picture.

FR06: The system shall provide different access permissions based on user roles.

FR07: The system shall allow users to log out.

4.2 Product Marketplace Requirements

FR08: Sellers shall be able to create product listings.

FR09: Sellers shall be able to edit product listings.

FR10: Sellers shall be able to remove product listings.

FR11: Buyers shall be able to browse available products.

FR12: Buyers shall be able to search for products.

FR13: Buyers shall be able to filter products based on available criteria.

FR14: Buyers shall be able to view product details.

FR15: Sellers shall be able to specify product type, price, quantity, selling method and location.

FR16: Sellers shall be able to upload product images.

FR17: Buyers shall be able to contact sellers.

FR18: Sellers shall be able to update product availability.

4.3 Workforce Marketplace Requirements

FR19: Workers and service providers shall be able to create service profiles.

FR20: Estate owners shall be able to search for workers and service providers.

FR21: The system shall categorize workers and services according to service type.

FR22: Workers shall be able to specify their charging method.

FR23: Estate owners shall be able to send job requests.

FR24: Workers shall be able to accept or reject job requests.

FR25: The system shall maintain the status of job requests.

4.4 Estate Management Requirements

FR26: Estate owners shall be able to register estates.

FR27: Estate owners shall be able to update estate information.

FR28: The system shall store estate information.

FR29: Authorized users shall be able to view estate information.

FR30: Estate owners shall be able to view estate related statistics.

4.5 Harvest Management Requirements

FR31: Authorized users shall be able to record harvest information.

FR32: The system shall store harvest records.

FR33: Authorized users shall be able to view harvest history.

FR34: The system shall maintain historical harvest information.

4.6 Rating and Review Requirements

FR35: Estate owners shall be able to rate workers after completed services.

FR36: Workers shall be able to rate estate owners after completed services.

FR37: Buyers shall be able to rate sellers after completed transactions.

FR38: Sellers shall be able to rate buyers after completed transactions.

FR39: Users shall be able to write reviews.

FR40: The system shall calculate and display average ratings.

4.7 Notification Requirements

FR41: The system shall notify users about new job requests.

FR42: The system shall notify users when job requests are accepted or rejected.

FR43: The system shall notify sellers when buyers contact them.

FR44: The system shall notify users about important account activities.

4.8 Administration Requirements

FR45: Administrators shall be able to view registered users.

FR46: Administrators shall be able to manage user accounts.

FR47: Administrators shall be able to verify users when required.

FR48: Administrators shall be able to monitor product listings.

FR49: Administrators shall be able to review reported content.

FR50: Administrators shall be able to suspend users who violate platform policies.

5. External Interface Requirements

5.1 User Interface

The system should provide a simple interface that can be understood by users with different levels of technical knowledge.

The interface will vary according to the user's role. Estate Owners, Estate Managers, Buyers, Sellers, Workers, Service Providers and Administrators will have access to different functions.

The main interfaces identified are the login and registration screens, dashboards, product marketplace, workforce marketplace, estate management, harvest management, profile, ratings and notifications.

5.2 Software Interfaces

The proposed software components are Flutter, Node.js, Express.js and PostgreSQL.

Firebase Cloud Messaging and Google Maps API may be integrated in future versions where required.

5.3 Communication Interfaces

The frontend will communicate with the backend using REST APIs.

Data exchanged between the frontend and backend will use JSON.

Secure communication will be implemented using HTTPS.

6. Non Functional Requirements

6.1 Performance

The system should provide reasonable response times during normal usage.

Search and filtering functions should return results without unnecessary delays.

6.2 Security

Passwords should be securely stored.

Protected functions should only be accessible to authenticated users.

Users should only be able to access functions permitted by their role.

Personal information should be protected from unauthorized access.

6.3 Usability

The system should be easy to understand and navigate.

The interface should use consistent navigation and terminology.

Error messages should be clear and understandable.

6.4 Reliability

The system should maintain the consistency of stored information.

The system should minimize the possibility of losing user and business data.

6.5 Scalability

The system should support the addition of new agricultural products and services.

New modules should be able to be introduced without major changes to the existing system.

6.6 Maintainability

The system should use a modular structure to make future maintenance easier.

The source code and technical documentation should be maintained throughout development.

7. Business Rules

BR01: Only registered users can access protected system functions.

BR02: Users can only access functions permitted by their assigned role.

BR03: Sellers must provide the required information before publishing a product listing.

BR04: Workers and service providers must provide their service category and charging method.

BR05: Estate owners should send a job request before hiring a worker through the platform.

BR06: Workers and service providers can accept or reject job requests.

BR07: Ratings and reviews should only be submitted after a completed transaction or service.

BR08: Estate managers should only manage estates assigned to them.

BR09: Administrators can suspend users who violate platform policies.

8. Future Enhancements

Possible future enhancements include online payment facilities, GPS based services, weather information, additional agricultural products, equipment rental, multilingual support, advanced analytics, AI based recommendations and estate tourism related services.

9. Document Status

This version represents the requirements after the initial requirements were organized into system modules. The requirements will continue to be reviewed and refined based on stakeholder feedback and development progress.